

Cooper Kimball-Rhines, Ph.D.*Data Science, Genomics, Conservation*

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<https://coopermkr.github.io>

I work with conservation practitioners and decision-makers to improve native plant restorations. My research is collaborative and interdisciplinary with a focus on translating genomic, spatial, and causal modeling research into restoration planning and policy.

Education

Ph.D., Environmental Biology, 2026

University of Massachusetts Boston

NSF Graduate Research Fellow

Dissertation: Adapting population genetics, practice, and policy for plant conservation in a changing world

Advisor: Brook Moyers

Committee: Mayra C. Vidal, Georgia Mavrommati, Alex Harkess, Bob F. Chen

B.Sc., Microbiology and Immunology, 2021

McGill University, School of Biomedical Sciences

Dean's Multidisciplinary Research List

Thesis: The PFAS package in *enviPath*: a resource for the prediction of AFFF biotransformation

Advisor: Jinxia Liu

Work Experience

Postdoctoral Research Associate, University of Massachusetts Amherst, June 2026-present

Leading USGS NECASC RISSC working groups on climate-smart restorations

Synthesizing knowledge on grassland plant genetics for restoration planning

Advisor: Bethany Bradley

Research Fellow, COBAL, June 2026-Sept. 2026

Geospatial data analyst for seagrass restoration planning in Casco Bay, Maine

Developed R Shiny tools to visualize field data for conservation decision-making

Pathways Intern, United States Environmental Protection Agency, Oct. 2021-Aug. 2024

EPA Student Intern of the Year 2023

Region 1 Superfund and Emergency Management Division

Assisted on-scene coordinators in assessing contamination at emergency removals

Manager: Stacy Greendlinger

Roadside Pollinator Assistant, New Hampshire Fish and Game, May-Sept. 2019-2021

Performed monarch butterfly surveys to inform NH DOT policy on habitat management

Supported Karner Blue Butterfly, Frosted Elfin, and White Mountain Fritillary projects

Manager: Heidi Holman

Publications

5. **Kimball-Rhines, C.**, Taveras Guzman, S., Mavrommati, G., Moyers, B. Leaping a conservation genetics gap: A collaborative, multidisciplinary approach to restoration genetics in the pine barren plant *Lupinus perennis*. In review *Conservation Science and Practice* <https://doi.org/10.64898/2026.08.31.748316>
4. **Kimball-Rhines, C.**, Moyers, B. Missing the forest for the trees: A review and case study of state conservation law. In press, *Oryx*.
3. Keller, A.G., McIntosh, T., **Kimball-Rhines, C.**, Leng, D., Nelson, K. 2026. Tension between opportunities and risks of deep learning in social-ecological systems research. *Ecology and Society* 31(3):6. <https://doi.org/10.5751/ES-17589-310306>
2. Comerford, M.S., Nash, K., Germaine, A.S., Andrews, C., Barr, T.S., Dain, J., Gonclaves, A., Jones, A., **Kimball-Rhines, C.**, Maguire, W., Richards, A., Schneider, E., Wilkes, A. and Vidal, M.C. (2026), Quantitative synthesis of the effects of drought on community composition and species interactions in terrestrial ecosystems. *Oikos* e12026. <https://doi.org/10.1002/oik.12026>
1. **Cooper Kimball-Rhines**, Alice E Kaisla, Scarlet Taveras Guzman, Emily Mello, Karol Vanessa Rojas Ramirez, Alex Harkess, Brook Moyers, The annotated, chromosome-scale *Salicornia depressa* (American pickleweed) genome, *Journal of Heredity*, 2025; esaf062, <https://doi.org/10.1093/jhered/esaf062>

Manuscripts In Prep

2. **Kimball-Rhines, C.***, Bock, B.*, Sarkar, M., Zahn, G. Graph neural networks reveal transferable structure for predicting community coalescence. In prep for *Nature Communications*
1. **Kimball-Rhines, C.**, Mello E., Taveras Guzman, S., Moyers B. Divergence in genotype and methylation are decoupled and are not driven by nitrogen source in a salt marsh plant. In prep for *Evolutionary Applications*

Research Grants and Fellowships

2023-2026	NSF Graduate Research Fellowship	\$159,000
2023-2025	Nantucket Biodiversity Initiative Climate Change and Island Resilience	\$5,506
2023-2025	DOE Joint Genome Institute Community Science Program RNA-seq Services	
2023	UMB Transdisciplinary Dissertation Proposal Development Program	\$3,000
2023	UMB Nancy Goranson Endowment Fund	\$735
2022	New England Botanical Society Graduate Student Research Award	\$1,500

Teaching Experience

Curriculum Maintainer, The Carpentries, Intro to R Geospatial Data, May 2025-present
 Instructor, iTCGA Computation for Cancer Genomics Workshop, June 2-20, 2025
 Instructor, The Command Line for Genomics Workshop, July 15-17, 2024
 Instructor, GIS in R for Casco Bay Workshop, Jan. 10-12, 2024
 Certified Instructor, The Carpentries, Dec. 2023-present
 Teaching Assistant, UMass Boston, Biostatistics and Experimental Design, Sept-Dec. 2023

Organized Workshops

- NE RISCC Plant Producer & Climate Adaptation Meeting, September 14, 2026, Online hosted by the Northeast Regional Invasive Species and Climate Change Network
- NE CASC Climate Adapted Restoration Meeting, April 9, 2026, Online hosted by the Northeast Regional Invasive Species and Climate Change Network

Invited Workshops

- ESIIL Leadership Program and Innovation Summit on AI for Sustainability: Translating Environmental Data into Decisions, May 11-14, 2026, Boulder, CO
- AIBS Policy Communication Bootcamp and Congressional Visits Day, April 28-30, 2025, Washington, DC
- NCEAS Environmental Data Science Summit: AI and Conservation Management, Feb. 4-6, 2025, Santa Barbara, CA

Media Features

2. Lambert, L. O. (2026, June 24) The Plight of the Karner Blue Butterfly. *National Wildlife Magazine*. Summer Issue. <https://www.nwf.org/Home/Magazines/National-Wildlife/2026/Summer/PhotoZone/Karner-Blue-Butterfly>
1. Brown, J. R. (2026) Vermont Lupine gets a Boost from Banked Seed. *Native Plant News*. Spring/Summer Issue. https://www.nativeplanttrust.org/documents/1821/NPT_loresSPREADS_NPNews.SprSum2026_28pg_1.pdf

Presentations

8. Poster: **Kimball-Rhines C.**, Taveras Guzman, S., Moyers B. “Gene Pool or Puddle: When ‘Local is Best’ Meets Restoration Reality.” Ecological Society of America 2025; August, 2025, Baltimore, MD
7. Poster: **Kimball-Rhines C.**, Moyers B. “Divergence and Diversity Decoupled: A Population Methyloomic Approach.” Harvard Plant Biology Initiative Meeting; May, 2025, Boston, MA.
6. Poster: **Kimball-Rhines C.** “Drowning to Diverse: A causal analysis of salt marsh sediment placement.” Northeast Estuarine Research Society Meeting; April, 2025, Provincetown, MA.
5. Oral Presentation: **Kimball-Rhines C.**, Palmer A., Harkess A., Guzman-Taveras S., Mello E., Rojas V., Moyers B. “Six Hot Tips learned from assembling a tetraploid halophyte genome.” Third Joint Congress on Evolutionary Biology; July, 2024, Montreal, QC, CA.
4. Oral Presentation: **Kimball-Rhines C.**, Palmer A., Moyers B. “Genetics for Marsh Conservation.” Nantucket Biodiversity Initiative Meeting; November, 2023, Nantucket, MA.
3. Seminar: **Kimball-Rhines C.**, Palmer A., Moyers B. “Live Demonstration of DNA Sequencing for Marsh Conservation.” Nantucket Field Station; August 15, 2023, Nantucket, MA.
2. Poster: **Kimball-Rhines C.**, Moyers B. “One Fish, Two Fish, Sea Fish, Bee Fish: Assessing State-Level Biodiversity Policy.” Society for the Study of Evolution; June, 2023, Albuquerque, NM.
1. Poster: **Kimball-Rhines C.**, Palmer A., Moyers B. “Genome Profiling of *Salicornia depressa*.” Ecological Society of America Meeting; August, 2023; Montreal, QC, CA.